Justo Darío Valverde Torres

Date of birth: 14/12/2005 | **Nationality:** Spanish, Peruvian | **Gender:** Male | **Phone:** (+34) 612293910 (Mobile) | **Email address:** justodariovalverde@gmail.com | **Website:** https://justodario.github.io | **LinkedIn:** https://www.linkedin.com/in/justodariovalverde/ | **Address:** Spain (Home)

EDUCATION & TRAINING

01/09/2023 - Current - MADRID, SPAIN

BSC ROBOTICS SOFTWARE ENGINEERING- UNIVERSIDAD REY JUAN CARLOS (URJC)

Robotics software architecture and development with strong emphasis on ROS 2–based systems and mobile robotics . Embedded and concurrent systems programming for real-time applications. Robot modeling and simulation using URDF,Gazebo and PyBullet, as well as mechanical design with Fusion360 and Blender. Machine Learning and Computer Vision. Planning and decision-making algorithms for cognitive robotic systems.

Level in EQF: 6

01/09/2025 - 03/02/2026 - LISBON, PORTUGAL

ERASMUS+ EXCHANGE PROGRAMME- INSTITUTO SUPERIOR TÉCNICO, UNIVERSIDADE DE LISBOA

During my Erasmus exchange in the third year of my Bachelor's degree, I took courses from the Master's programme in Electrical and Computer Engineering at IST.

Level in EQF: 7

04/07/2026 - 20/07/2026 - CHENGDU, CHINA

UESTC 2026 SMEE INTERNATIONAL ROBOTICS SUMMER SCHOOL- UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA (UESTC)

WORK EXPERIENCE

15/01/2025 - CURRENT - MADRID, SPAIN

SCARA PROJECT LEADROBOTECH URJC

Lead a student team developing a 4-DOF SCARA robotic arm.

Implemented forward and inverse kinematics to achieve precise joint and end-effector control.

Combining precise control, computer vision and decision algorithms, a demo was developed in Arduino in which the

SCARA was able to play Connect-3. Demo.

Currently migrating the software stack to ROS2 + MoveIt and designing a suitable end-effector in order to scale the

project to an autonomous chess-playing robot.

01/06/2025 - CURRENT - MADRID, SPAIN

ASSOCATION TREASURERROBOTECH URJC

Managed budget and purchasing decisions for the association's projects.

Supported new project teams by advising on component selection related to cost/performance trade-offs.

25/10/2025 - 15/07/2026 - LISBON, PORTUGAL

AUTONOMOUS SYSTEMS - MAPPING LEAD TÉCNICO SOLAR BOAT

Responsible for the mapping implementation within a ROS2 autonomous navigation stack for the SP01 vessel, aiming

to prepare it for the Njord Challenge.

Built an initial OccupancyGrid mapping pipeline using LIDAR point clouds as the input.

Currently designing a new GridMap-based, costmap-style mapping pipeline, using GridMap layered architecture to

combine multiple 2D map representations into a single map composed of the sum of the following layers:

- OccupancyGrid based dynamic layer using log-odds and Bayesian updates.
- Inflation layer inspired by ROS costmaps for safety margins around obstacles.
- Obstacle layer that uses buoy positions received from an AI perception node(camera + LIDAR fusion), to constrain the navigable space into a corridor between left and right buoys, tailored for the Njord Challenge requirements.

Latest recorded test: LINK .

01/06/2025 - CURRENT

PROJECT 1 – ROS2 QUADRUPED ROBOT

Built a 12 DOF quadruped robot from scratch, covering electronics, 3D design and printing and a complete ROS2 control architecture.

Identified hardware limitations in the first design iteration, including an off-centered center of mass and servo

backlash/free-play, causing actuation uncertainty and errors in path following.

Currently developing a reactive control pipeline to address the mentioned issue and drafting a paper based on this

work.

Preparing robust locomotion as a preliminary step before autonomous navigation and LLM based decision making and human-robot interaction.

Presented the project at the Europe Embodied Summit 2026 in Munich.

Link: https://justodario.github.io/#projects/0/0

01/06/2024 - 19/09/2024

PROJECT 2 - SEMI-AUTONOMOUS RC CAR

Built a remote control car as a hands-on entry project to learn electronics, 3D design and 3D printing fundamentals.

DEMO

MADRID, SPAIN

HACKATHONS

Europe Embodied 48H Hackathon Munich - Jun 2026

I was selected as one of the 100 participants of the hackathon. Participated in the WendyOS Challenge, in which I worked alongside my team to grant a Unitree Go2 robot dog autonomous factory inspection capabilities and human robot interaction.

Hackathon AIA Image classification model fine tunning and atari boxing agent training - Apr 2026

My team participated in the Image Classification challenge and the Atari Boxing Agent tournament, in which I got to apply Machine Learning and Reinforcement Learning.

URJC Hackathon 2025(Mapfre Challenge) - Jul 2025

Developed a complementary mental-health care mobile app concept for Mapfre clients.

Qubic x Vottun Web3 Hackathon - Mar 2025

Built a web app designed to make crypto transfers more accessible, by enabling users to send cryptocurrency by

scanning a QR code and entering an amount through a simplified wallet connected interface that made use of smart

contracts.

OpenRed Hackathon - Feb 2025

Developed a desktop application that, when a Radiacode-102 device is connected, gamma radiation measurements

are read and in addition to location estimated from IP triangulation are then uploaded to the OpenRed database.

I enjoy participating in hackathons in my free time because they push me to find solutions fast, develop my team working and leading skills

and give me the opportunity to improve my ability of pitching/presenting technical ideas under pressure.

SKILLS

English(C1) | Spanish(Native) | French (basic)

Robotics Software
ROS, ROS2 | Experience with Robot Operating System (ROS) and Linux (ROS1 and ROS2) | C++ based on ROS 1 and ROS2 | MicroROS

Programming and Embedded Systems
Knowledge of FreeRTOS | C | Python | Java | C++

Perception and ML
OpenCV | YOLO | Computer Vision | PointCloud and Laserscan | Pytorch/Tensorflow (Basic ML model training)

Control and Locomotion
ros2_control | Reinforcement Learning(Basics) | PID | Inverse Kinematics | Forward Kinematics

Simulation, modeling and CAD
URDF | XACRO | Gazebo | Pybullet | Blender | Fusion360 | use CAD software

Tools & Workflows
Git | Linux | Foxglove | Docker | Rviz

Mapping, Navigation and Planning
Nav2 | PlanSys2 | Gridmap | OccupancyGrid | A* | Costmap | Mapping Algorithms

LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C1

DRIVING LICENCE

Driving licence: B